FITBIT USER BEHAVIOR DATA ANALYSIS REPORT

TO: URŠKA SRŠEN & BELLABEAT EXECUTIVE TEAM
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SUBJECT: GUIDING BELLABEAT RELAUNCH W/ FITBIT DATA
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SUMMARY

Bellabeat is preparing to relaunch its brand and reposition its main product IVY+ as a premium health tracker designed specifically for women. An analysis of Fitbit user data showed that smart device wearers are predominantly sedentary, and the bulk of their activity is of low intensity. After a digital presence audit, it is this report's suggestion that Bellabeat revamp its website, relaunch its social media accounts, and present the IVY+ as a capable health tracker with menstruation tracking capabilities- deploying triggers to current and prospective users during observed activity peaks.

BELLABEAT STRATEGIC POSITIONING

Bellabeat's market position has remained focused on delivering women value through specialized smart devices to track menstruation and general health. It has been unable to gain a significant portion of smart device market share with incumbents like Fitbit, Apple, and Samsung continuing to dominate the broader market. In response, Bellabeat has narrowed its product portfolio to the IVY+ and its companion subscription service, concentrating its offering around the one dimension where it meaningfully differentiates: correlating menstrual cycle data, lifestyle habits, and biometric readings to deliver a comprehensive picture of women's health.

This narrowing has come alongside a significant reduction in social media presence and a restriction of product sales to Bellabeat's own direct channel- both of which raise the barrier to discovery for new customers. Rebuilding awareness will require meeting the target market where their existing wearable device habits already are. Understanding how women currently use devices like Fitbit will identify the utility sought by current wearable device users and allow Bellabeat to effectively communicate the depth of its product.

ANALYSIS

Of all distance categories compared against total steps, light active distance and very active distance returned the strongest correlations. Both show a positive relationship with step count. (see Fig. 1) Most users cluster at lower step counts with a small number of outliers at the high end, pointing to predominantly casual behavior across the dataset. Activity follows consistent patterns in intensity across the week. (see Fig. 2) Intensity rises outside of sleeping hours, with clear weekday peaks between 5–8PM after traditional working hours. Weekends show higher overall intensity, but with peaks shifting into the late morning and early afternoon. Users in this dataset spent 81.96% of their 24-hour day sedentary, with the remaining 18.04% being active. (see Fig. 3) Of that active time, 84.86% is spent on light activities and only 9% on very active movement (see Fig. 4).

STRATEGIC CONTEXT

Activity patterns from this dataset suggest health consciousness over fitness prioritization- users are tracking their bodies, not training them. This distinction reveals how Bellabeat should position its value proposition. IVY+ is a tool to help women understand and improve their daily rhythms without demanding behavioral overhaul. Weekday evenings, when activity intensity peaks and health is most likely to be front of mind, represent the most receptive window for reaching both prospective and current users. This information can be further utilized when modifying the behavior of the AI Coach.

DATA PROCESS

Fitbit data was obtained through Kaggle, uploaded into a Postgres database, and queried through DBeaver. Full documentation of data manipulation and validation is available in this project's data log.

DATA LIMITATIONS

Data used in this analysis is from March 12th to May 12th of 2016 and contains no more than 35 users across all datasets. The age of data is not a large concern as there is no evident reason to believe that smart device users' behavior has significantly changed in a way that would make this analysis invalid. The biggest gap in information is the exclusion of age and gender of users, meaning findings could be misrepresentative of Bellabeat's target market behavior.

ANALYSIS NEXT STEPS

Gathering a larger, more recent dataset limited to women would capture insights most in line with Bellabeat's target market. A 385-person sample size that includes strictly women aged 18-35 would yield more reliable insights, meeting the standard threshold for a 95% confidence level with a 5% margin of error across a large consumer market. Because it is central to the product's value proposition, gathering telemetry from menstruation tracking apps would add a dimension of insight into relevant use cases.

Findings from this analysis suggest wearable device users tend toward moderate, routine activity rather than high-intensity exercise. Cross referencing sleep and activity location data would reveal context about the types of activities users are participating in- whether they are exercising at home, in a gym, or during commutes. Richer lifestyle context would allow Bellabeat to better communicate with its target market.

RECOMMENDATIONS

Based on Bellabeat's current market standing and the insights observed from the Fitbit user data, re-establishing Bellabeat's market presence with an increased focus on overall lifestyle tracking is the strongest course of action. Bellabeat should continue to highlight its prioritization of menstrual cycle tracking without allowing it to overshadow its broader lifestyle benefits. This strategy sets up Bellabeat customers to purchase Bellabeat's AI Coach once they have passed through the funnel.

The following are advised:

1. Revive digital presence by relaunching on platforms most active among women 18–35, such as Instagram and TikTok, and reintroducing paid digital ads to address the discovery gap created by reduced distribution and social channels.
2. Add additional distribution channels like Amazon to lower barriers to entry for new customers.
3. Send lifestyle tracking messaging to current and prospective users during times when health is likely to be front of mind- weekdays from 5-8PM and late morning to early afternoon on weekends.

DATA VISUALIZATIONS

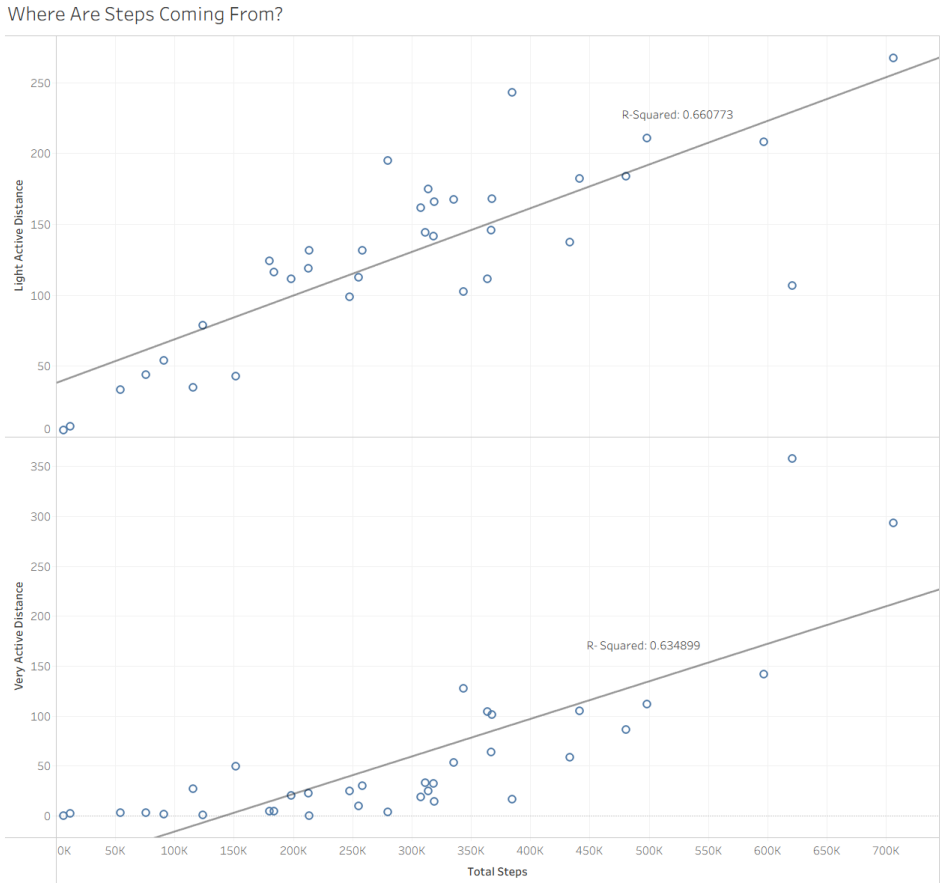

Figure 1: Where Are Steps Coming From? - The overall distance traveled correlates most significantly with light and very active activity.

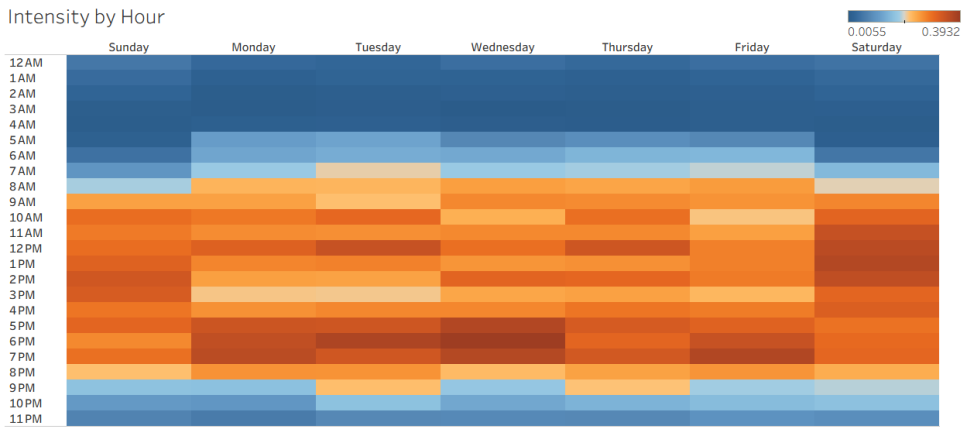

Figure 2: Intensity by Hour- Users' behavior on weekdays follows a clear pattern of mild intensity throughout the day with spikes occurring after traditional working hours from 5-8pm. On the weekends, intensity is more evenly spread, with intensity peaking in the late morning to early afternoon.

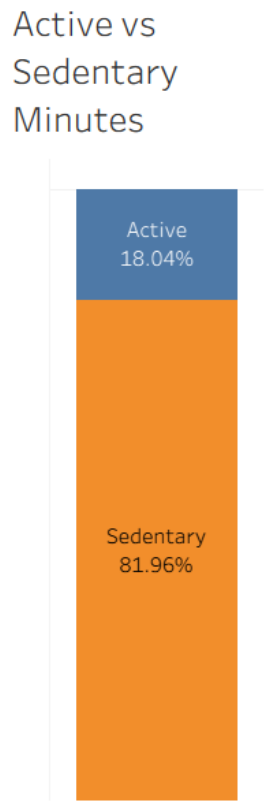

Figure 3: Active vs Sedentary Minutes- FitBit users spend the majority of their time sedentary, with active minutes accounting for only 18.04% of a 24-hour day.

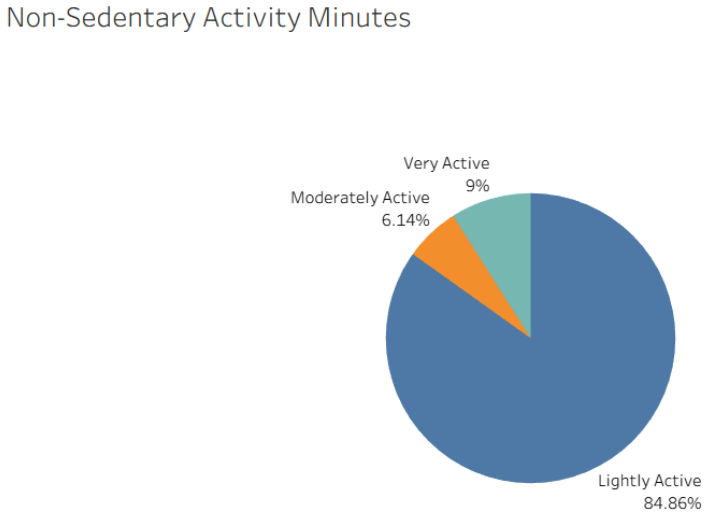

Figure 4: Non-Sedentary Activity Minutes- Of the activity recorded, lightly active time takes up 84.86% of all time spent active. Very active time is only 1.62%, 9% of 18.04%, of a 24-hour day.